

SAMPLE LESSON · CASE STUDY: THE ENERGY TRANSITION

Can a City Cut Carbon Without Going Broke?

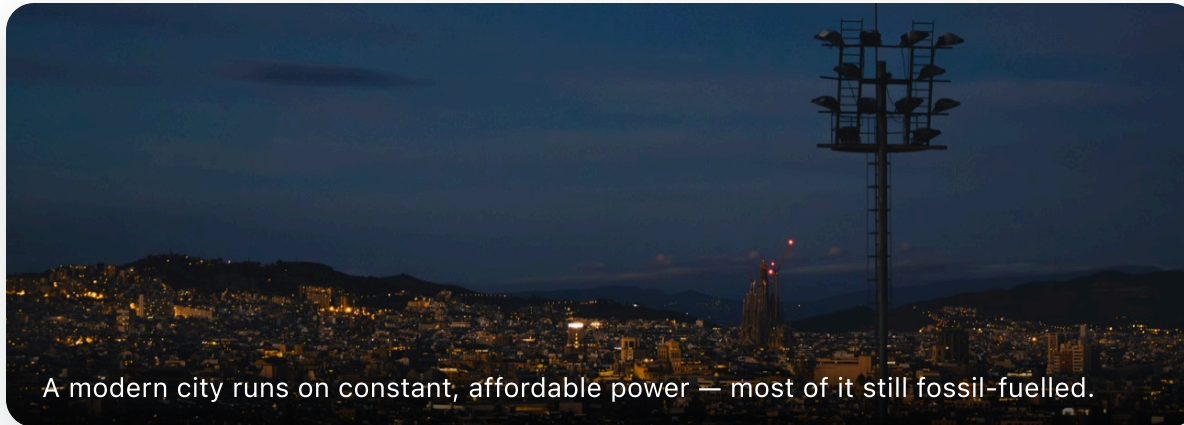
A look inside a World Science Academy research session — where a real sustainability dilemma becomes award-winning research.



TODAY'S CHALLENGE

The climate says cut emissions now.

Families and factories say keep energy cheap and reliable. Can we do both?



A modern city runs on constant, affordable power — most of it still fossil-fuelled.

BY THE END, STUDENTS CAN

- ✓ Compare energy sources on cost, carbon, and reliability
- ✓ Explain why clean energy isn't automatically the cheap choice
- ✓ Use real data to recommend a smart policy mix

THE CARBON PROBLEM

Why this is hard to fix

Most of the world's electricity still comes from burning fossil fuels — and the CO₂ it releases warms the planet for centuries.

- 01 Fossil-heavy grid**
Around 60% of global electricity still burns coal and gas.
- 02 Long-lived CO₂**
Carbon dioxide lingers in the atmosphere for centuries.
- 03 Rising demand**
EVs, AI, and growth keep pushing electricity use up.



Burning fossil fuels for power is the single largest source of CO₂ emissions.

Every grid juggles three goals

Affordable

Energy people and businesses can actually pay for.

Reliable

Power that is there whenever it is switched on.

Clean

Low carbon and low pollution over its whole life.

The hard part: **no single source wins on all three at once.**

KEY CONCEPTS

The language students build

Carbon emissions (CO₂e)

Greenhouse gases released, measured in carbon-dioxide equivalent.

Levelized cost (LCOE)

The all-in cost to make one unit of electricity over a source's lifetime.

Intermittency

Sun and wind aren't always available — their output rises and falls.

Carbon price

A fee on emissions that makes polluting cost money.



Photovoltaic cells turn sunlight straight into electricity — clean, but only when the sun shines.

Students do, not just listen



Seeing the real thing — a working solar array — makes the data mean something.

In-Person

- Real-time polling & word clouds
- Think-Pair-Share discussions
- Live dataset & graph analysis
- Policy role-play debates

Online

- Social annotation of case studies
- Micro-module self-checks
- Small-group discussion pods
- Energy-mix simulations

No source wins on every measure

Students weigh cost, carbon, and reliability across five ways to make electricity.

Source	Cost (LCOE, \$/MWh)	Carbon (g CO ₂ /kWh)	Reliability
Wind	\$38	11	Variable
Solar PV	\$40	41	Variable
Nuclear	\$70	12	Very high
Natural gas	\$60	490	High
Coal	\$90	820	High

Illustrative teaching figures (order-of-magnitude, based on public LCOE & lifecycle-emissions ranges).

THE BIG IDEA

There is no free lunch — only trade-offs

Wind and solar are cheap and clean but **intermittent**. Gas and coal are steady but **dirty**. Nuclear is clean and reliable but **slow and costly to build**. The winning answer is rarely one source — it is a **smart mix plus policy**.

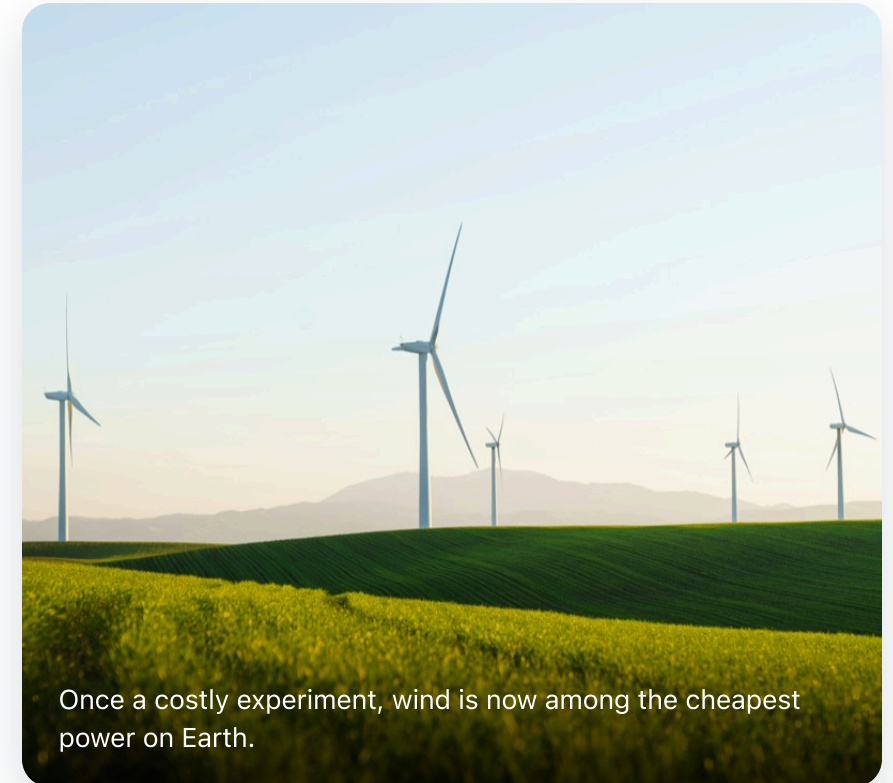
Clean ≠ automatically cheap

Reliability has a cost

The answer is a portfolio

The cost of clean energy collapsed

- 2010 Solar ~\$380/MWh — far too costly to compete.
- 2015 Costs fall ~80% as manufacturing scales worldwide.
- 2015 Paris Agreement — nations pledge to cut emissions.
- 2020 Solar & wind become the **cheapest new power** in most markets.
- 2022 Big subsidies (e.g. US IRA) **accelerate** the build-out.
- Today The bottleneck shifts from cost to **storage & the grid**.



Once a costly experiment, wind is now among the cheapest power on Earth.

Turning this into a science-fair study

Ask a testable question

- Does a carbon price cut emissions faster than subsidies?
- How much storage does a 50%-solar grid need?
- Does rooftop solar actually pay off in our city?

Design a fair test

- Define the variable & use real public datasets
- Model cost vs. carbon, not just one number
- Graph the trade-off — then argue from it

Our role: we coach students through each step — framing the question, gathering credible data, modelling the trade-offs, and presenting like a real researcher.

WHAT STUDENTS WALK AWAY WITH

Deliverables

01

A research question

A focused, testable hypothesis they can defend.

02

Data & model

Real datasets, a clear method, and honest graphs.

03

Poster & talk

Competition-ready board and a confident presentation.

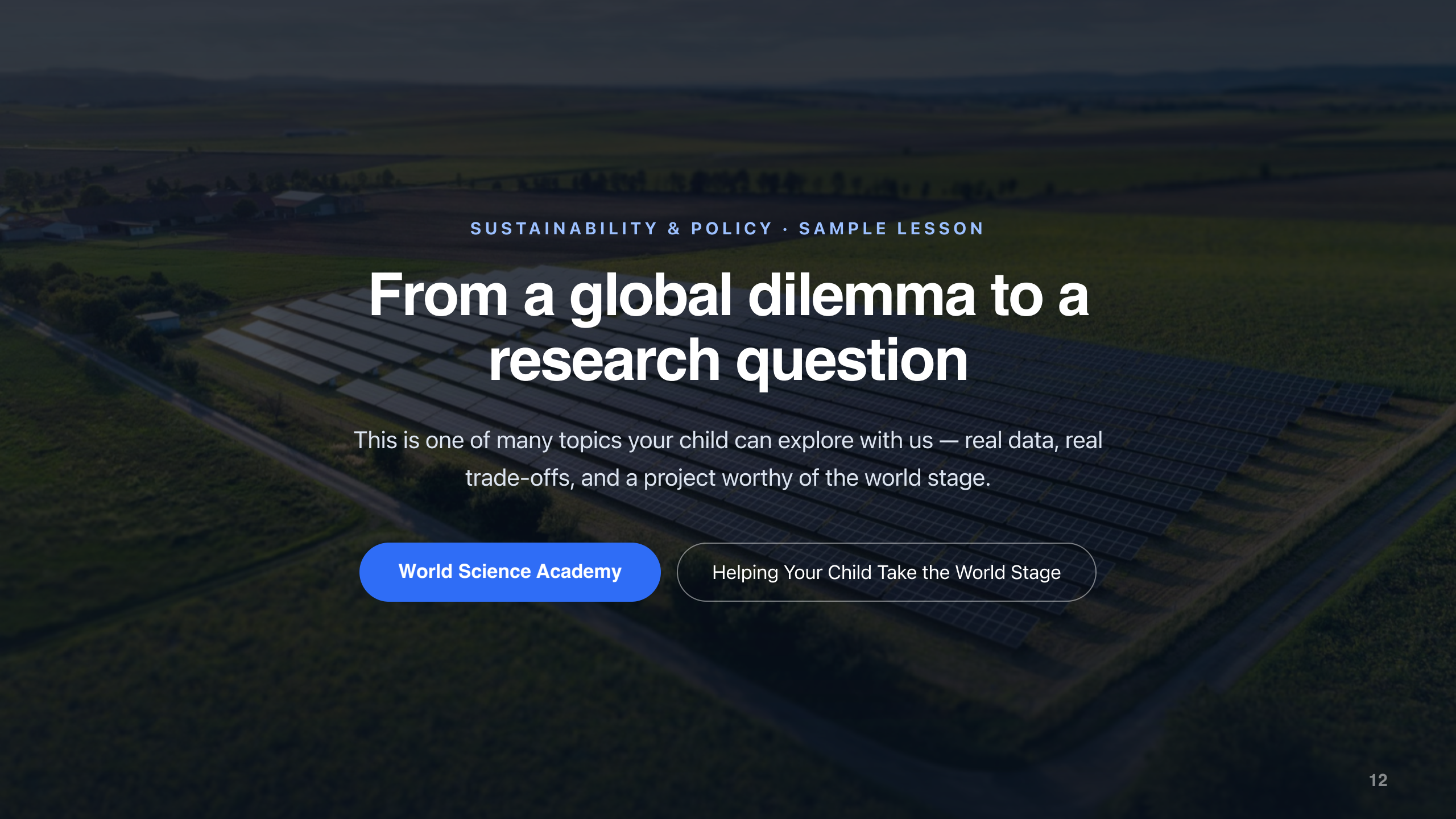
A policy recommendation

A data-backed argument for a specific energy or policy mix.

Judge-ready Q&A

Practice fielding tough questions like a researcher.



An aerial photograph of a large solar farm with rows of solar panels installed in a field. In the background, there are rolling hills and a small village with houses and trees. The image is darkened to serve as a background for the text.

SUSTAINABILITY & POLICY · SAMPLE LESSON

From a global dilemma to a research question

This is one of many topics your child can explore with us — real data, real trade-offs, and a project worthy of the world stage.

World Science Academy

Helping Your Child Take the World Stage